RoadSense.ai – Task 4: Data Analytics

Preliminary analysis of the 2022 road-accident working dataset

1. Objective

Inspect the working dataset's structure, summarize existing data-status labels, and prepare review files to identify records requiring source and geographic-level verification.

2. Variables used

Year, State/UT, District, Accident Cases, Injured, Died, Data Status, and Source. Record ID and Data Quality Note are also present where available. The dataset does not provide verified weather, road-condition, vehicle-type, or exact-coordinate fields, so no conclusions are drawn about those factors.

3. Initial dataset checks

Metric	Value
Dataset rows	759
Distinct State/UT labels	35
Distinct district names	755
Missing State/UT values	0
Missing District values	0
Missing Accident Cases	0
Missing Injured values	0
Missing Died values	0
Exact duplicate rows	0

These checks describe structural completeness and exact duplication only. They do not independently verify source accuracy or prove that every row represents a district-level observation.

4. Existing data-status labels

Existing label group	Record count
Official State/UT data	576
Estimated	123
Records carrying a Verified label (not independently confirmed)	60

These are selected label-group counts, not a complete mutually exclusive breakdown. The word 'Verified' in a status field is not independent proof that every metric in a row has been checked against its source.

5. Current analytics outputs

The safer analytics script creates descriptive summaries and review files. It intentionally does not publish a top-district ranking because independent source verification is incomplete.

- Data quality summary: structural checks, label counts, and review indicators.
- Data status counts: counts for each existing Data Status label.
- District record counts by State/UT: row counts, not accident totals.
- Source verification review: row-level review outcomes and geographic-level warnings.

- Estimated district records: rows carrying the Estimated label.
- Verified-label review: rows carrying a Verified label that still require source checking.
- District records for source check: candidate records to compare with reliable district-level sources before any ranking.

6. Data quality, limitations, and interpretation

- Data Status and Source are text labels; their presence alone does not independently prove a record is accurate.
- Records labelled Official State/UT data must not be treated as district-level observations unless the underlying source confirms that geographic level.
- Records labelled Estimated must not be presented as directly reported, verified district statistics.
- Repeated accident metrics across multiple district rows are a signal for source review, not conclusive proof of an error by themselves.
- Do not sum district rows to claim official State/UT totals unless the records are complete, mutually exclusive, and supported by the source.
- Tamil Nadu and Haryana records whose source fields name official publications remain candidates for metric-by-metric checking; matching source text is not independent verification.
- The dataset lacks verified coordinates, so a reliable geographic accident-hotspot map cannot be produced from this CSV alone.

7. Conclusion

The Task 4 script ran successfully and generated descriptive summaries and source-review files without modifying the original Task 3 cleaned CSV. The current outputs are suitable for preliminary data-quality review. Confirmed district rankings should only be produced after relevant district-level figures have been independently checked against reliable sources.

8. Files generated

- RoadSense_AI_Task_4_Data_Analytics_Safer.py
- RoadSense_AI_Task_4_Data_Quality_Summary.csv
- RoadSense_AI_Task_4_Data_Status_Counts.csv
- RoadSense_AI_Task_4_District_Record_Counts_by_State.csv
- RoadSense_AI_Task_4_Source_Verification_Review.csv
- RoadSense_AI_Task_4_Estimated_District_Records.csv
- RoadSense_AI_Task_4_Verified_Label_Review.csv
- RoadSense_AI_Task_4_District_Records_For_Source_Check.csv

9. Reproduction

Place RoadSense_AI_Task_4_Data_Analytics_Safer.py beside RoadSense_AI_Task_3_Cleaned_Road_Accident_Data_2022.csv and run:

python RoadSense_AI_Task_4_Data_Analytics_Safer.py